

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES	
10							***** MODULAR VALVES *****			812	NIPPLE TO BE CUT FROM SS SCH XXS PIPE.
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.
20	U04		MODULAR DBB	K15AC02	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0		USE ONLY FOR TIE IN
30	U04		MODULAR DBB	K15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR MODELLING PURPOSE AT TIE-IN POINT.DIMENSIONAL DATA NOT CONFIRMED.
40	U04	VAV	MODULAR DBB	K09AC06	3	3	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 CL.1 316 SS TRIM METAL LO FULL SIZE TYPE 2BV	0	0	825	ALL BOLTS ARE METRIC DIA.
50	U04		MODULAR DBB	M15AC02	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B1	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10
60	U04		MODULAR DBB	M15AC06	2	2	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1 INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22
70	U04	VAW	MODULAR DBB	M15EC02	0.75	1	MODULAR VALVE DB&B BALL TYPE 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE R2BVR	1.5	2	B6	TO BE USED WITH RAISED FACE FLANGES ONLY.
80	U04	VAW	MODULAR DBB	M09EC02	1	1	MODULAR VALVE DB&B BALL TYPE 900# X 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	BA1	TO BE USED WITH COMPACT FLANGES REQUIRING STANDARD NUTS ONLY.
90	U04		MODULAR DBB	K09AC07	8	8	MODULAR VALVE DB&B BALL TYPE 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT GO FULL BORE TYPE 2BV	0	0	BA4	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.
100	U04	817	MODULAR DBB	K09CC07	14	14	MODULAR VALVE DB&B BALL TYPE 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT GO FULL BORE TYPE 2BV	0	0	FA8	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES
110							***** BALL VALVES *****			FF5	USE FLANGES, GASKETS & VALVES AS REQUIRED TO MATCH EQUIPMENT OR VESSEL NOZZLES.
110		VB2								FS1	ONLY TO BE USED FOR CONNECTION BETWEEN CARBON STEEL AND CLAD LINE CLASSES
120	U04		BALL	A15AC05	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT RB LO	0	0	GA4	SPECIAL GASKET ONLY FOR USE AS SPECIFIED IN ENG-200-117
130	U04		BALL	A15AC06	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL RB LO	0	0	SB6	USE JACKSCREWS FOR PADDLE BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.
140	U04		BALL	A09AC05	3	3	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB LO	0	0	UD7	USE THREDOLET AND SCREWED PLUG FOR HYDROSTATIC TEST VENTS AND DRAINS. SEAL WELD AFTER HYDROSTATIC TEST
150	U04		BALL	A09AC06	3	3	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB LO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.
160	U04		BALL	A09AC07	4	10	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0	UJ3	USE 3IN WITH THERMOWELL CONNECTIONS ONLY
170	U04		BALL	A09AC08	4	10	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0	UJ4	DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.
180	U04	817	BALL	A09CC07-K21	12	36	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT RB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
190	U04	817	BALL	A09CC08-K21	12	36	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL RB GO	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
200	U04		BALL	B15AC01	0.75	1.5	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM FLOATING SOFT FB LO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
210	U04		BALL	B15AC02	0.75	1.5	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM FLOATING METAL FB LO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
220	U04	VA5	BALL	B15AC05	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION SOFT FB LO	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
230	U04	VA5	BALL	B15AC06	2	2	BALL 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM TRUNNION METAL FB LO	0	0	VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.
240	U04	VA5	BALL	B09AC05	3	3	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB LO	0	0		
250	U04	VA5	BALL	B09AC06	3	3	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB LO	0	0		
260	U04	VA5	BALL	B09AC07	3	10	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
270	U04	VA5	BALL	B09AC08	3	10	BALL 900# RF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
280	U04	817	BALL	B09CC07-K21	12	36	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION SOFT FB GO	0	0		
290	U04	817	BALL	B09CC08-K21	12	36	BALL 900# CF LTCS A350 LF2 CL.1 316SS TRIM TRUNNION METAL FB GO	0	0		
300	U04	825, 817	BALL	BE5C2	2	2	BALL 1500# RTJ LTCS A350 LF2 CL.1 316SS TRIM FS FB LONG PATT	0	0		
310							***** THROUGH CONDUIT GATE VALVES *****			UK9	DO NOT USE FOR ORIFICE FLANGE TAP ASSEMBLIES.
310										VA5	USE THESE VALVES ONLY WHERE SPECIFIED ON PIPING AND INSTRUMENT DIAGRAMS
320			GATE	W09AC70	3	4	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB HW	0	0	VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.
330			GATE	W09AC71	6	10	GATE 900# RF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.
340	U04	817	GATE	W09CC71	12	36	GATE 900# CF LTCS A352 LCC 316 SS TRIM THROUGH COND SOLID SLAB GO	0	0	VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.
350							***** GLOBE VALVES *****			VC1	SPRING PISTON CHECK VALVES ARE PREFERED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.
350										VQ2	QUARTER TURN VALVES ARE TO BE SELECTED FOR PROCESS APPLICATIONS.
360											
360	U04		GLOBE	L15AC48	0.75	1.5	GLOBE 1500# RF LTCS A350 LF2 CL.1 INCONEL OS&Y HW	0	0		
370	U04		GLOBE	L15AC41	2	2	GLOBE 1500# RF LTCS A352 LCC INCONEL OS&Y HW	0	0		
380			GLOBE	L09AC41	3	3	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y HW	0	0		
390			GLOBE	L09AC42	4	8	GLOBE 900# RF LTCS A352 LCC TRIM 16 OS&Y GO	0	0		
400							***** CHECK VALVES *****				
400											
410											
410	U04	VC1	CHECK	C15AC50	0.75	1.5	CHECK PISTON 1500# RF LTCS A350 LF2 CL.1 INCONEL TRIM	0	0		
420	U04		CHECK	C15AC45	2	2	CHECK DUAL WAFER LUG 1500# RF LTCS A352 LCC INCONEL TRIM THROU BOLTED	0	0		
430	U04		CHECK	C09AC45	3	8	CHECK DUAL WAFER LUG 900# RF LTCS A352 LCC TRIM 16 THROU BOLTED	0	0		
440	U04		CHECK	C09AC52	10	10	CHECK DUAL DBL FLG 900# RF LTCS A352 LCC TRIM 16	0	0		
450	U04	817	CHECK	C09CC52-K21	12	36	CHECK DUAL DBL FLG 900# CF LTCS A352 LCC TRIM 16	0	0		
460							*****PIPE*****				
460											
470											
470	U04		PIPE	5358512	0.5	1.5	PIPE SCH 160 SMLS LTCS A333 GR.6 PE	0	0		
480			PIPE	5358515	2	3	PIPE SCH 160 SMLS LTCS A333 GR.6 BE	0	0		
490			PIPE	5358510	4	16	PIPE SCH 120 SMLS LTCS A333 GR.6 BE	0	0		
500			PIPE	5358600	18	24	PIPE SCH 120 EFW LTCS A671 CC60 CL.22 BE	0	0		
510			PIPE	5358627	30	30	PIPE WT 50.0MM EFW LTCS A671 CC60 CL.22 BE	0	0		
520			PIPE	5358625	36	36	PIPE WT 59.0MM EFW LTCS A671 CC60 CL.22 BE	0	0		

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 260 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 1 of 4	900K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
530		812	PIPE	5358511	0.75	0.75	PIPE SCH XXS SMLS LTCS A333 GR.6 PE	0	0	
540							***** FITTINGS *****			
550										
550			ELBOW	5392816	0.75	3	ELL BW 90 DEG LR SCH 160 LTCS A420 WPL6	0	0	
560	U04	UK9	ELBOW	5392815	0.75	0.75	ELL BW 90 DEG LR SCH XXS LTCS A420 WPL6	0	0	
570			ELBOW	5392700	4	16	ELL 90 DEG LR SCH 120 LTCS A420 WPL6	0	0	
580			ELBOW	5392388	18	24	ELL 90 DEG LR SCH 120 LTCS A420 WPL6-W	0	0	
590	U04		ELBOW SR	5392245	24	24	ELL BW 90 DEG SR SCH 120 LTCS A420 WPL6-W	0	0	
600			ELBOW	5392547	30	30	ELL 90 DEG LR WT 50.0MM LTCS A420 WPL6-W	0	0	
610			ELBOW	5392538	36	36	ELL 90 DEG LR WT 59.0MM LTCS A420 WPL6-W	0	0	
620			ELBOW	5392822	0.75	3	ELL BW 45 DEG LR SCH 160 LTCS A420 WPL6	0	0	
630			ELBOW	5392701	4	16	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6	0	0	
640			ELBOW	5392392	18	24	ELL 45 DEG LR SCH 120 LTCS A420 WPL6-W	0	0	
650	U04	UK9	ELBOW	5392831	0.75	0.75	ELL BW 45 DEG LR SCH XXS LTCS A420 WPL6	0	0	
660			ELBOW	5392549	30	30	ELL 45 DEG LR WT 50.0MM LTCS A420 WPL6- W	0	0	
670			ELBOW	5392548	36	36	ELL 45 DEG LR WT 59.0MM LTCS A420 WPL6-W	0	0	
680										
680			TEE	5392825	0.75	3	TEE BW SCH 160 LTCS A420 WPL6	0	0	
690	U04	UK9	TEE	5392879	0.75	0.75	TEE SCH XXS LTCS A420 WPL6	0	0	
700			TEE	5392702	4	16	TEE BW SCH 120 LTCS A420 WPL6	0	0	
710			TEE	5392404	18	24	TEE BW SCH 120 LTCS A420 WPL6-W	0	0	
720			TEE	5392417	30	30	TEE BW WT 50.0MM LTCS A420 WPL6- W	0	0	
730			TEE	5392416	36	36	TEE BW WT 59.0MM LTCS A420 WPL6-W	0	0	
740			TEE RED	5392821	0.75	2	TEE RED BW SCH 160 LTCS A420 WPL6	1	3	
750			TEE RED	5392470	1.5	3	TEE RED SCH 120/160 LTCS A420 WPL6	4	6	
760			TEE RED	5392603	4	14	TEE RED SCH 120 LTCS A420 WPL6	6	16	
770			TEE RED	5392878	8	20	TEE RED SCH 120 LTCS A420 WPL6-W	18	24	
780			TEE RED	5392872	12	24	TEE RED WT 50.0MM/SCH 120 LTCS A420 WPL6-W	30	30	
790			TEE RED	5392835	20	24	TEE RED WT 59.0MM/SCH 120 LTCS A420 WPL6-W	36	36	
800										
800			CAP	5392826	0.75	3	CAP BW SCH 160 LTCS A420 WPL6	0	0	
810			CAP	5392703	4	24	CAP SCH 120 LTCS A420 WPL6	0	0	
820			CAP	5392627	30	30	CAP WT 50.0MM LTCS A420 WPL6	0	0	
830			CAP	5392628	36	36	CAP SCH 59.0MM LTCS A420 WPL6	0	0	
840										
840	U04		RED CONC	5392827	0.5	2	REDUCER CONC SCH 160 LTCS A420 WPL6	1	3	
850			RED CONC	5392467	1.5	3	REDUCER CONC SCH 120/160 LTCS A420 WPL6	4	6	
860			RED CONC	5392704	4	14	REDUCER CONC SCH 120 LTCS A420 WPL6	6	16	
870			RED CONC	5392707	10	20	REDUCER CONC SCH 120 LTCS A420 WPL6-W	18	24	
880			RED CONC	5392314	24	24	REDUCER CONC WT 59.0MM/SCH 120 LTCS A420 WPL6-W	36	36	
890			RED CONC	5392332	20	24	REDUCER CONC WT 50.0MM/SCH 120 LTCS A420 WPL6-W	30	30	
900			RED CONC	5392334	30	30	REDUCER CONC WT 59.0MM/50.0MM LTCS A420 WPL6-W	36	36	
910	U04		RED ECC	5392828	0.5	2	REDUCER ECC SCH 160 LTCS A420 WPL6	1	3	
920			RED ECC	5392468	1.5	3	REDUCER ECC SCH 120/160 LTCS A420 WPL6	4	6	
930			RED ECC	5392705	4	14	REDUCER ECC SCH 120 LTCS A420 WPL6	6	16	
940			RED ECC	5392709	10	20	REDUCER ECC SCH 120 LTCS A420 WPL6-W	18	24	
950			RED ECC	5392317	24	24	REDUCER ECC WT 59.0MM/SCH 120 LTCS A420 WPL6-W	36	36	
960			RED ECC	5392333	20	24	REDUCER ECC WT 50.0MM/SCH 120 LTCS A420 WPL6-W	30	30	
970			RED ECC	5392335	30	30	REDUCER ECC WT 59.0MM/50.0MM LTCS A420 WPL6-W	36	36	
980										
980							***** OLETS *****			
990										
990	U04		OLETS	5504170	0.75	2	NIPOFLANGE 1500# RF SCH 160 LTCS A350 LF2 CL.1	2	3	
1000	U04		OLETS	5504171	0.75	2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2 CL.1	4	24	
1010	U04		OLETS	5504172	0.75	2	NIPOFLANGE 1500# RF 50.0MM/SCH 160 LTCS A350 LF2 CL.1	30	30	
1020	U04		OLETS	5504173	0.75	2	NIPOFLANGE 1500# RF 59.0MM/SCH 160 LTCS A350 LF2 CL.1	36	36	
1030	U04	UJ2	OLETS	5504163	1.5	1.5	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2 CL.1	4	24	
1040	U04	UJ2	OLETS	5504174	1.5	1.5	NIPOFLANGE 1500# RF 50.0MM/160 LTCS A350 LF2 CL.1	30	30	
1050	U04	UJ2	OLETS	5504175	1.5	1.5	NIPOFLANGE 1500# RF 59.0MM/160 LTCS A350 LF2 CL.1	36	36	
1060	U04	UJ4	OLETS	5504156	2	2	NIPOFLANGE 1500# RF 50.0MM/160 LTCS A350 LF2 CL.1	30	30	
1070	U04	UJ4	OLETS	5504154	2	2	NIPOFLANGE 1500# RF 59.0MM/160 LTCS A350 LF2 CL.1	36	36	
1080	U04	UJ4	OLETS	5504366	2	2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2 CL.1	10	10	
1090	U04	UJ4	OLETS	5504366	2	2	NIPOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2 CL.1	18	24	
1100	U04	UJ3	OLETS	5505324	3	3	NIPOFLANGE 900# RF SCH 120/160 LTCS A350 LF2 CL.1	24	24	
1110	U04	FF5	OLETS	5504321	0.75	0.75	NIPOFLANGE FLAT BOTTOM 1500# RF SCH 160 LTCS A350 LF2 CL.1	0	0	
1120										
1120	U04		OLETS	5505392	0.75	1	WELDOLET SCH 160 LTCS A350 LF2 CL.1	2	3	
1130	U04		OLETS	5505394	0.75	3	WELDOLET SCH 120/160 LTCS A350 LF2 CL.1	4	24	
1140	U04		OLETS	5505625	0.75	3	WELDOLET WT 50.0MM/SCH 160 LTCS A350 LF2 CL.1	30	30	
1150	U04		OLETS	5505613	0.75	3	WELDOLET WT 59.0MM/SCH 160 LTCS A350 LF2 CL.1	36	36	
1160	U04		OLETS	5505195	4	8	WELDOLET SCH 120 LTCS A350 LF2 CL.1	12	24	
1170	U04		OLETS	5505193	4	10	WELDOLET WT 50.0MM/SCH 120 LTCS A350 LF2 CL.1	30	30	

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		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 260 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 2 of 4	900K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1180	U04		OLETS	5505612	4	12	WELDOLET WT 59.0MM/SCH 120 LTCS A350 LF2 CL.1	36	36	
1190										
1190	U04	UJ2	ELBOFLG	5516082	1.5	1.5	ELBOFLANGE 1500# RF SCH 120/160 LTCS A350 LF2 CL.1	4	24	
1200	U04	UJ2	ELBOFLG	5516083	1.5	1.5	ELBOFLANGE 1500# RF 50.0MM/160 LTCS A350 LF2 CL.1	30	30	
1210	U04	UJ2	ELBOFLG	5516084	1.5	1.5	ELBOFLANGE 1500# RF 59.0MM/160 LTCS A350 LF2 CL.1	36	36	
1220										
1220	U04		ELBOFLG	5509804	1.5	1.5	ELBOLET WT 50MM/SCH 160 LTCS A350 LF2 CL.1	30	30	
1230										
1230	U04	UD7	OLETS	5514772	0.75	1	THREDOLET 3000# LTCS A350 LF2 CL.1	2	36	
1240										
1240		816					***** FLANGES *****			
1250										
1250	U04	FA8	TEE RED	5599704	0.5	0.5	FLG SCRDR (FNPT) REDUCING 1500# RF LTCS A350 LF2 CL.1	0.75	1.5	
1260										
1260			FLANGE	5851205	0.75	0.75	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
1270			FLANGE	5851205	2	2	FLANGE ADAPTER 1500# RF 316 SS X 12 MM TUBE	0	0	
1280										
1280	U04		FLANGE	5557397	0.5	2	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
1290	U04	UK9	FLANGE	5557505	0.75	0.75	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH XXS BORE	0	0	
1300	U04	817	FLANGE	5557511	0.75	2	FLG WN 1500# RTJ LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
1310	U04	817		5557283	4	4	FLG WN 900# RTJ LTCS A350 LF2 CL.1 SCH 120 BORE	0	0	
1320	U04	817		5557283	8	8	FLG WN 900# RTJ LTCS A350 LF2 CL.1 SCH 120 BORE	0	0	
1330	U04	817		5557281	3	3	FLG WN 900# RTJ LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
1340	U04		FLANGE	5557145	3	3	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 160	0	0	
1350	U04		FLANGE	5557143	4	10	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 120 BORE	0	0	
1360	U04		FLANGE	5557143	12	24	FLG WN 900# RF LTCS A350 LF2 CL.1 SCH 120 BORE	0	0	
1370	U04		FLANGE	5556494	30	30	FLG WN 900# RF LTCS A350 LF2 CL.1 WT 50.0MM B16.47 SERIES A	0	0	
1380	U04		FLANGE	5556495	36	36	FLG WN 900# RF LTCS A350 LF2 CL.1 WT 59.0MM B16.47 SERIES A	0	0	
1390	U04		FLANGE	5557395	18	18	FLG WN 1500# RF LTCS A350 LF2 CL.1 SCH 120 BORE	0	0	
1400	U04	817	FLANGE	5619821	12	24	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 SCH 120 BORE	0	0	
1410	U04	817	FLANGE	5619822	30	30	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 WT 50.0MM	0	0	
1420	U04	817	FLANGE	5619823	36	36	FLG COMPACT 900# RJ LTCS A350 LF2 CL.1 WT 59.0MM	0	0	
1430	U04		BLIND	5615616	0.5	2	FLG BLIND 1500# RF LTCS A350 LF2 CL.1	0	0	
1440	U04		BLIND	5615605	3	10	FLG BLIND 900# RF LTCS A350 LF2 CL.1	0	0	
1450	U04		BLIND	5615605	12	24	FLG BLIND 900# RF LTCS A350 LF2 CL.1	0	0	
1460	U04		BLIND	5621082	30	36	FLG BLIND 900# RF LTCS A350 LF2 CL.1 B16.47 SERIES A	0	0	
1470	U04	817		5615614	0.75	1.5	FLG BLIND 1500# RTJ LTCS A350 LF2 CL.1	0	0	
1480	U04	817	BLIND	5615614	2	2	FLG BLIND 1500# RTJ LTCS A350 LF2 CL.1	0	0	
1490	U04	817	BLIND	5615606	3	4	FLG BLIND 900# RTJ LTCS A350 LF2 CL.1	0	0	
1500	U04	817	BLIND	5615606	8	8	FLG BLIND 900# RTJ LTCS A350 LF2 CL.1	0	0	
1510	U04	817	BLIND	5619773	12	36	FLG COMPACT BLIND 900# RJ LTCS A350 LF2 CL.1 LINE CLASS 900K21	0	0	
1520										
1520							***** ORIFICE FLANGES *****			
1530										
1530	U04		ORIFICE FLG	5624598	2	2	WN ORIF SET 1500# RF LTCS A350 LF2 CL.1 SCH 160 BORE	0	0	
1540			ORIFICE FLG	5624537	3	3	WN ORIF SET 900# RF LTCS A350 SCH 160 BORE	0	0	
1550	U04		ORIFICE FLG	5624532	4	24	WN ORIF SET 900# RF LTCS A350 SCH 120 BORE	0	0	
1560	U04	817	ORIFICE FLG	5624538	12	24	WN ORIF SET COMPACT 900# RJ LTCS A350 SCH 120 BORE	0	0	
1570	U04		ORIFICE FLG	5624367	36	36	WN ORIF SET 900# RF LTCS A350 LF2 CL.1 59.0MM WT 796.4MM BORE ASME B16.47 SERIES A	0	0	
1580										
1580							***** PIPE NIPPLES *****			
1590										
1590		812		0			NIPPLE BBE - CUT FROM PIPE			
1600										
1600							***** GASKETS *****			
1610										
1610			GASKET	5672251	0.5	2	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1620		FS1	GASKET	5672251	6	6	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1630		FS1	GASKET	5672251	18	18	GASKET 1500# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1640	U04	GA6	GASKET	5673021	2	2	GASKET 1500# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
1650			GASKET	5673117	3	24	GASKET 900# 316SS SPIRAL WOUND GRAPHITE IR	0	0	
1660			GASKET	5674033	30	36	GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
1670	U04	GA6	GASKET	5673020	3	3	GASKET 900# 316SS KAMMPROFILE GRAPHITE ASME B16.5	0	0	
1680	U04	GA4	GASKET	5674035	36	36	GASKET 900# ALLOY 825 KAMMPROFILE GRAPHITE ASME B16.47 SERIES A	0	0	
1690	U04	GA4	GASKET	5673700	18	24	GASKET 900# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
1700	U04	817	GASKET	5673443	12	36	GASKET 900# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 900K21	0	0	
1710	U04	817	GASKET	5674479	0.75	2	GASKET 1500# RTJ SS OCTAGONAL RING	0	0	
1720	U04	817	GASKET	5674481	3	4	GASKET 900# RTJ SS OCTAGONAL RING	0	0	
1730	U04	817	GASKET	5674481	8	8	GASKET 900# RTJ SS OCTAGONAL RING	0	0	
1740										
1740		B1					***** BOLTING *****			
1750										
1750	U04		BOLTING	5675770	0.5	20	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 260 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 3 of 4	900K21

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1760	U04	BA4	BOLTING	5675770	24	30	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1770	U04	BA4	BOLTING	5675776	36	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
1780	U04	BA1	BOLTING	5675776	24	99	STUD-BOLT WITH 2 NUTS API-6A718	0	0	
1790										
1790		SB6	BOLTING	5675675	10	36	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1800										
1800							***** BLINDS AND SPACERS *****			
1810										
1810			SPEC BLIND	5749595	0.75	2	SPECTACLE BLIND 1500# RF LTCS A516 GR.70	0	0	
1820			SPEC BLIND	5749596	3	8	SPECTACLE BLIND 900# RF LTCS A516 GR.70	0	0	
1830	U04	825	SPEC BLIND	5749589	2	2	SPECTACLE BLIND 1500# RTJ FEMALE LTCS A516-GR70	0	0	
1840	U04	825, 817		5749591	8	8	SPECTACLE BLIND 900# RTJ FEMALE LTCS A516-GR70	0	0	
1850										
1850			PADDLE	5750102	10	10	PADDLE BLANK 900# RF LTCS A516 GR.70	0	0	
1860			PADDLE	5750102	12	24	PADDLE BLANK 900# RF LTCS A516 GR.70	0	0	
1870			PADDLE	5750202	10	10	PADDLE SPACER 900# RF LTCS A516 GR.70	0	0	
1880			PADDLE	5750202	12	24	PADDLE SPACER 900# RF LTCS A516 GR.70	0	0	
1890			PADDLE	5750052	30	36	PADDLE BLANK 900# RF LTCS A516-GR70 B16.47 SERIES A	0	0	
1900			PADDLE	5750254	30	36	PADDLE SPACER 900# RF LTCS A516-GR70 B16.47 SERIES A	0	0	
1910	U04	817	PADDLE	5750218	12	36	PADDLE BLANK COMPACT 900# RJ LTCS A516 GR.70	0	0	
1920	U04	817	PADDLE	5750224	12	36	PADDLE SPACER COMPACT 900# RJ LTCS A516 GR.70	0	0	
1930										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT - ALL SIZES SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 3 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS FLANGE RATING: 900 RF/CF TEMPERATURE RANGE: -40 / 260 °C COMMENT: REFER NOTE 800	12/16/2024	BY OZAC	U04	Page 4 of 4	900K21